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Comparative Analysis of Iterative and Direct Methods for Solving Electrical Network Linear Systems

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Abstract

This study investigates the performance of three iterative numerical methods, the Jacobi Method, the Gauss-Seidel Method, and the Successive Over-Relaxation (SOR) Method in solving linear systems derived from electrical network models. Electrical networks are mathematically represented as systems of linear equations based on Kirchhoff's laws, with node voltages as unknown variables. Efficient numerical techniques are essential for solving these systems, particularly for networks where direct analytical solutions are difficult or impractical.

In this research, a dataset representing a small electrical network is converted into a matrix system of the form $Ax = b$. The Jacobi, Gauss–Seidel, and SOR methods are implemented in Python to compute approximate solutions for the unknown node voltages. A direct solver is used as the reference solution to evaluate the accuracy of the iterative methods. The SOR method is further analyzed for different relaxation parameters (ω) to identify the optimal value that minimizes error and iteration count. The algorithms are evaluated based on convergence rate, number of iterations, and approximation error, with convergence behavior tracked for each node across iterations.

Experimental results demonstrate that the Gauss-Seidel method generally converges faster than Jacobi due to its use of updated values within each iteration. The SOR method, when using an optimized relaxation parameter, can achieve even faster convergence and lower error than both Jacobi and Gauss-Seidel. The findings highlight the relative strengths and limitations of each iterative approach for practical numerical computation in electrical network analysis.

This study contributes to understanding the effectiveness of iterative numerical techniques in solving real-world engineering problems, providing insights into method selection and parameter tuning for accurate and efficient computation.

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1 Introduction

Electric networks are fundamental components of modern technological architecture that supports power distribution, communication systems, and electronic devices. The analysis of these networks often involves solving linear equations from circuit laws such as Kirchhoff's Current Law. As electrical systems become more complex, analytical methods become inefficient or impractical for solving large systems of equations. Numerical Analysis offers approaches by applying computation algorithms to approximate solutions efficiently. According to recent research on generalized iterative methods, numerical linear algebra plays a critical role in moving from manual analytical solving to computational methods for large matrices (Nonlaopon et al., 2023).

Iterative methods such as Jacobi, Gauss–Seidel, and Successive Over-Relaxation (SOR) are widely used for solving systems of linear equations due to their computational efficiency and lower memory requirements, particularly when dealing with large and sparse matrices. In electrical engineering applications, circuit equations can be expressed in matrix form, allowing both iterative and direct solver methods to determine unknown voltages and currents within the network. Direct methods provide exact solutions in a finite number of steps. However, they often require higher computational cost and memory, making them less suitable for large-scale systems. In contrast, iterative methods generate approximate solutions through repeated computations and are more efficient for complex networks with limited resources. Studies involving large-scale systems such as memristor crossbar arrays emphasize the advantage of iterative techniques in handling strict memory constraints (Machado et al., 2024). Since these methods differ in terms of convergence speed, accuracy, and computational efficiency, a comparative analysis is necessary. Therefore, this study aims to evaluate and compare the performance of direct solver methods and iterative techniques, specifically Jacobi, Gauss–Seidel, and Successive Over-Relaxation in solving electrical network linear systems.

2 Preliminaries

This section presents the fundamental concepts and computational techniques necessary for the analysis of electrical network linear systems. It provides the mathematical background and numerical frameworks that support the application of both direct and iterative methods in solving systems of equations arising from electrical circuit analysis. These preliminaries establish the foundation for understanding how different solution techniques operate and how their performance can be evaluated in terms of accuracy, convergence, and computational efficiency.

2.1 Fundamentals of Numerical Analysis

Numerical analysis is a field of applied mathematics that focuses on developing algorithms to approximate solutions to mathematical problems that cannot be solved easily using analytical formulas. Many real-world engineering problems involve complex equations that require computational methods for practical solutions.

Iterative computational methods have become essential in engineering for efficiently resolving systems of linear equations derived from complicated mathematical models (Nonlaopon et al., 2023). In electrical circuit analysis, numerical algorithms help determine currents in circuits by solving systems of equations from circuit laws. These algorithms rely on mathematical structures such as matrices and vectors, which represent relationships between variables in a compact form. Recent studies focusing on mixed-precision algorithms have emphasized that numerical methods play a vital role in resolving large, sparse matrices safely and efficiently during simulations (Fucci et al., 2022). Because electrical networks often produce large systems of linear equations, numerical techniques provide efficient computational strategies for analyzing circuit behavior and system performance.

2.1.1 System of Linear Equations

A system of linear equations consists of multiple equations containing the same set of variables where each equation represents a linear relationship between the variables. These systems appear frequently in scientific and engineering problems, particularly in modeling physical systems. In electrical circuit analysis, systems of linear equations arise when applying Kirchhoff's laws to determine currents within a network. The shift from manual analytical solving to computational iterative methods has been highlighted as absolutely necessary for managing these complex linear relationships (Nonlaopon et al., 2023). As the number of equations increases, solving the system manually becomes impossible, especially for large electrical networks containing many interconnected components. Computational algorithms are therefore used to approximate solutions efficiently. Modern research in multidomain modeling has shown that linear systems derived from engineering models can involve thousands of variables, making numerical techniques essential for solving them without overwhelming system memory (Giampiccolo et al., 2021).

2.1.2 Matrix Representation of Linear System

Matrix representation provides a compact and efficient way of expressing systems of linear equations. In matrix form, a system can be written as $Ax = b$, where A represents the coefficient matrix, x represents the vector of unknown variables, and b represents the constant vector. Matrix representation simplifies the computational process because algorithms can operate directly on them, which is particularly useful when handling sparse approximate factorizations in massive systems (Fucci et al., 2022). In electrical circuit analysis, the coefficient matrix often represents conductance relationships between nodes in the circuit, while the vector of unknown variables represents currents. By converting the circuit equations into matrix form, computational algorithms can efficiently perform operations such as matrix multiplication and iterative updates. Recent research highlights that matrix-based numerical algorithms are widely used in engineering simulations because they enable scalable computation for large systems, such as complex deep-learning hardware architectures (Machado et al., 2024).

2.1.3 Iterative Methods

Iterative methods are numerical techniques used to approximate solutions to systems of equations by repeatedly improving an initial estimate. Instead of computing the exact solution in a single step, iterative algorithms update the solution gradually until it converges to an acceptable level of accuracy. According to recent evaluations of iterative methods for dense linear systems, these techniques are particularly useful for solving large systems because they require significantly less memory than direct factorizations (Wang et al., 2021).

These methods are widely used in engineering simulations, computational physics, and optimization problems. Each iteration calculates new values for the unknown variables based on previous approximations and the structure of the coefficient matrix. The convergence of the algorithm depends on factors such as the properties of the matrix and the chosen initial values. Recent studies utilizing incomplete-LU preconditioned methods have demonstrated that iterative techniques remain essential tools in modern circuit simulation applications (NVIDIA, 2021). Among the most commonly used iterative methods for solving linear systems are the Jacobi and Gauss–Seidel algorithms.

2.2 Electric Network Modeling

Electrical network modeling involves representing electrical circuits as mathematical systems that describe the relationships between currents. This modeling process allows engineers to analyze circuit behavior without physically constructing the system. Mathematical models derived from electrical circuit laws provide the fundamental baseline for analyzing these systems. Research into the simulation of large-scale arrays highlights that computational modeling of electrical networks is increasingly necessary as physical architectures become more dense and complex (Machado et al., 2024). By applying Kirchhoff's laws to each node or loop in the circuit, engineers can derive a system of equations that describes the behavior of the entire network. This mathematical representation enables the efficient multidomain modeling of highly nonlinear electromagnetic circuits (Giampiccolo et al., 2021).

2.2.1 Kirchhoff's Current Law

Kirchhoff's Current Law (KCL) states that the total current entering a node in an electrical circuit must equal the total current leaving the node. This law is based on the principle of conservation of electric charge and is widely used in circuit analysis. When applying KCL to a circuit, each node generates an equation representing the balance of currents flowing into and out of that node. By writing these equations for every node in the network, a system of linear equations is formed. In complex networks, the number of nodes can be massively scaled, resulting in a system of equations that strictly requires computational techniques for an efficient solution. Recent applications of iterative methods to intuitionistic fuzzy systems show that KCL-based nodal analysis remains a fundamental approach for evaluating voltage inputs and outputs in modern circuit modeling (Saw et al., 2023).

2.2.2 Formation of Electrical Network Linear System

The formation of electrical network linear systems involves converting the relationships between electrical components into mathematical equations. One of the most commonly used approaches is nodal analysis, which applies Kirchhoff's Current Law to each node in the circuit. Nodal analysis is a highly efficient method for generating linear equations, which can then be optimized using advanced iterative techniques like SOR or Gauss–Seidel for much faster convergence (Saw et al., 2023). In this approach, each node voltage is treated as an unknown variable, and equations are written to describe the current relationships between nodes. These equations are then arranged into matrix form to produce a system of linear equations representing the entire circuit. The resulting matrix system can be solved using numerical algorithms to determine the unknown voltages.

2.3 Direct Solver Method

Direct solver methods are numerical techniques that compute the exact solution of a system of linear equations through a finite sequence of mathematical operations. Common examples include Gaussian elimination and LU decomposition. While direct methods are reliable and produce accurate solutions for well-conditioned matrices, recent studies on array simulations note that direct solvers become computationally expensive and face severe memory constraints when dealing with very large electrical networks (Machado et al., 2024). Furthermore, multidomain modeling research emphasizes that the computational intensity of direct solvers for large circuit equations often necessitates a complete shift toward iterative relaxation methods in modern engineering (Giampiccolo et al., 2021). Therefore, direct solver methods currently serve best as useful reference solutions when checking the baseline accuracy of iterative algorithms.

2.4 Successive Over-Relaxation

The Successive Over-Relaxation (SOR) method is an extension of the Gauss–Seidel iterative technique designed to accelerate convergence. In this method, a relaxation factor is introduced to control the rate at which new approximations are updated. According to recent studies on determining optimal relaxation parameters, choosing the correct relaxation factor can drastically improve the convergence speed of these iterations (Liu, 2021). The SOR method is widely used in numerical simulations because it can reduce the overall computational load. Recent advancements highlight the efficacy of third-refinement techniques in SOR methods to further accelerate this convergence (Audu et al., 2024). Moreover, improved iterative schemes using SOR have been proven to increase stability and actively decrease the spectral radius of linear systems derived from differential models (Zaheer Ahmed et al., 2022).

2.5 Jacobi Method

The Jacobi method is one of the simplest iterative algorithms used to solve systems of linear equations. In this method, each variable is updated using the values from the previous iteration, meaning that all variables are updated simultaneously. According to modern comparative studies on iterative methods, the Jacobi method is widely used as a foundational technique primarily because of its conceptual simplicity and ease of implementation (Wang et al., 2021). The algorithm begins with an initial guess for the solution and repeatedly updates each variable until the approximation converges. Because the method does not use updated values within the same iteration, it may require many iterations to converge. However, the Jacobi method remains highly relevant in specialized applications, such as preconditioned iterative methods utilized in parallelized, GPU-accelerated circuit simulations (NVIDIA, 2021).

2.6 Gauss Seidel Method

The Gauss–Seidel method is another iterative technique used to solve systems of linear equations. Unlike the Jacobi method, the Gauss–Seidel algorithm immediately uses newly computed values within the same iteration. This sequential approach often leads to significantly faster convergence, making it highly effective for specific electrical applications, including solving voltage inputs in fuzzy linear systems (Saw et al., 2023). Because of this improvement in convergence speed, the Gauss–Seidel method is widely favored in numerical simulations over the standard Jacobi approach. Recent research confirms that Gauss–Seidel-based algorithms remain a critical benchmark when evaluating the performance and efficiency of any new iterative method proposed for dense linear systems (Wang et al., 2021).

3 Application of Iterative and Direct Methods

This section presents the application of both direct and iterative methods in solving electrical network linear systems. It focuses on how these computational techniques are utilized to determine unknown variables such as node voltages and branch currents derived from circuit equations. The implementation of each method is discussed in the context of matrix-based representations of electrical networks, highlighting their procedural approaches and computational behavior. Through these applications, the study demonstrates how different methods perform when applied to the same system, providing a basis for evaluating their efficiency, accuracy, and practicality in real-world scenarios.

3.1 Overview

Electrical network problems are commonly formulated as systems of linear equations using fundamental circuit laws. These equations are typically expressed in matrix form as $Ax = b$, where A represents the coefficient matrix derived from circuit parameters, x is the vector of unknown variables, and b is the vector of known constants. Solving this system efficiently is essential for accurately analyzing circuit behavior. In this study, both direct and iterative methods are applied to solve the resulting system of equations. The direct solver method computes the exact solution in a finite number of steps, making it suitable for smaller systems or cases where high precision is required.

In contrast, iterative methods including Jacobi, Gauss–Seidel, and Successive Over-Relaxation (SOR) approximate the solution through repeated calculations, gradually improving accuracy with each iteration. The application process involves initializing values, then applying the chosen algorithm, and finally evaluating convergence based on a predefined tolerance level. Each method is implemented on the same electrical network model to ensure consistency in comparison. Key performance indicators such as convergence speed, number of iterations, computational efficiency, and accuracy are considered in assessing the effectiveness of each method, providing a framework for the detailed analysis presented in the succeeding subsections.

3.2 Application of Direct Solver Method

The direct solver computes the exact node voltages by solving the system of linear equations $Ax = b$, where A is the conductance matrix with the reference node removed and b is the current injection vector. The solution provides the voltage of each node relative to the reference node or Node 5. The direct solver serves as the reference solution for evaluating iterative methods.

Node	Voltage (V)
1	0.876098
2	0.745873
3	0.602449
4	0.712935
5	0.000000

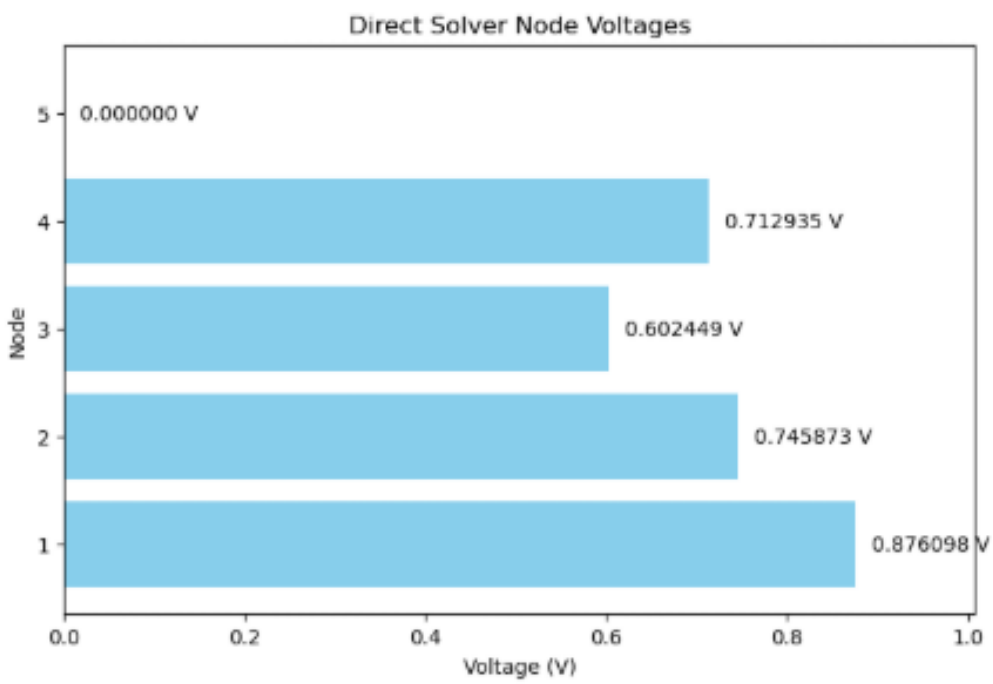

Figure 1: Direct Solver Node Voltages

3.3 Application of Successive Over-Relaxation

SOR is an iterative method that improves Gauss–Seidel by introducing a relaxation factor ω (omega). This factor accelerates convergence by partially updating the node voltages with a weighted combination of the new and previous values. In this study, $\omega = 1.1$ is used to test the convergence speed and error relative to the direct solver.

Present Table (Example for $\omega = 1.25$)

Node	Voltage (V)
1	0.876098
2	0.745873
3	0.602449
4	0.712935

Optional Table: SOR performance for different ω.

Omega	Iterations	Error
1.05	24	7.15×10^{-7}
1.10	21	6.82×10^{-7}
1.25	13	1.11×10^{-7}
1.50	23	2.60×10^{-7}

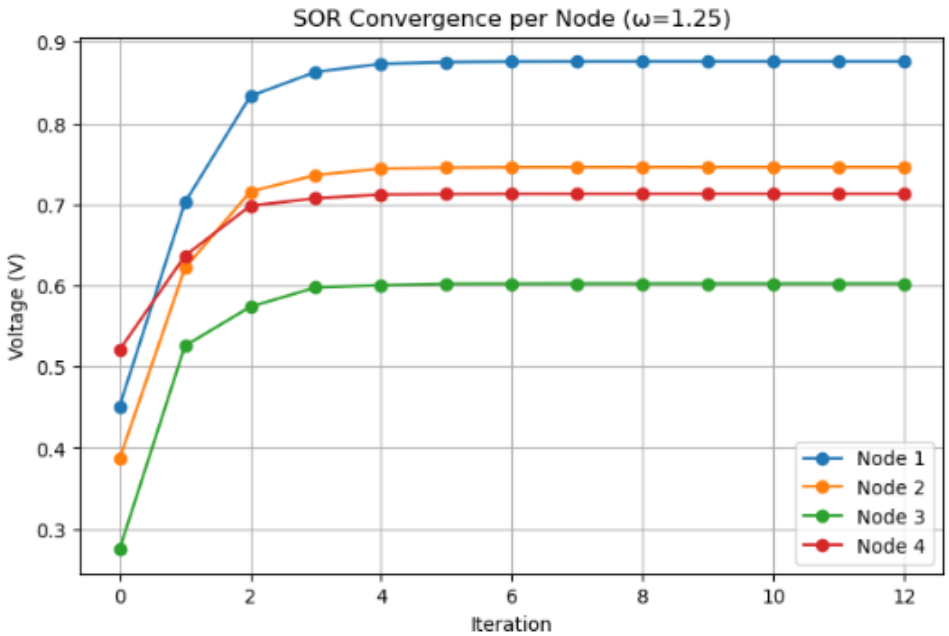

Figure 2: SOR Convergence Per Node

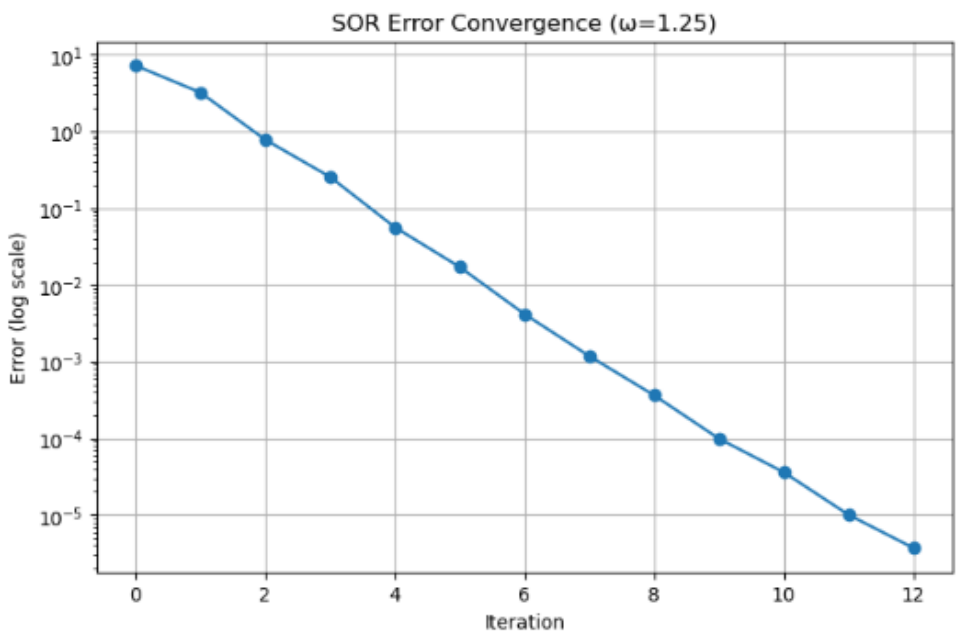

Figure 3: SOR Error Convergence

3.4 Application of Jacobi

Jacobi is a method where all node voltages are updated simultaneously using values from the previous iteration. It is simple but usually slower than Gauss–Seidel and SOR because it does not use newly updated values within the same iteration.

Node	Voltage (V)
1	0.876095
2	0.745870
3	0.602446
4	0.712932

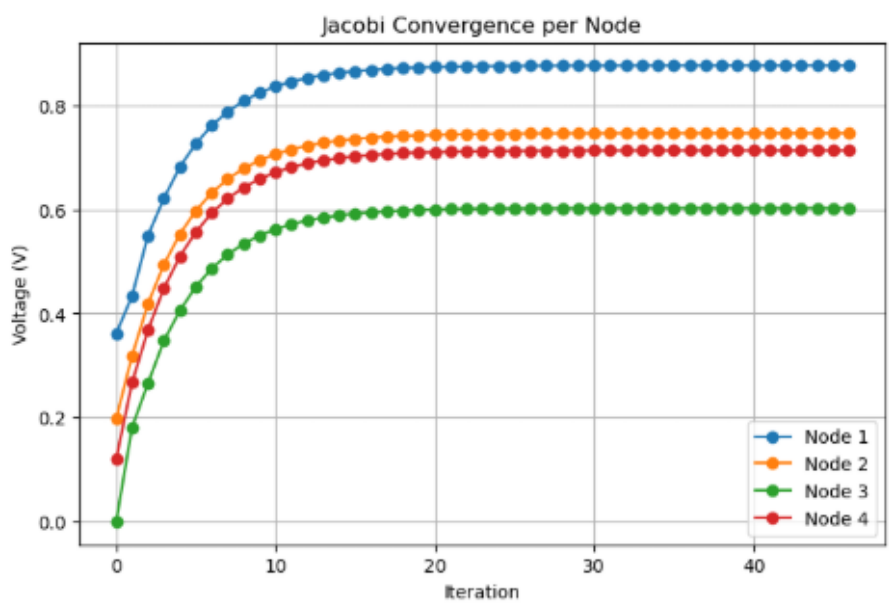

Figure 4: Jacobi Convergence Per Node

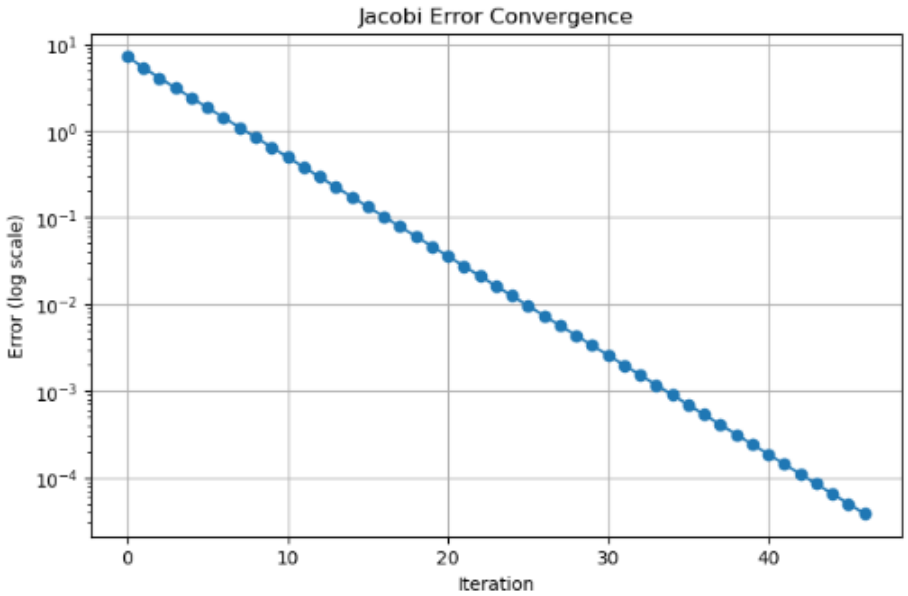

Figure 5: Jacobi Error Convergence

3.5 Application of Gauss-Seidel

Improving Jacobi, Gauss–Seidel updates the node voltages in place, meaning the latest values are immediately used in the calculation of subsequent nodes in the same iteration. This typically results in faster convergence than Jacobi.

Node	**Voltage (V)**
1	0.876096
2	0.745872
3	0.602448
4	0.712934

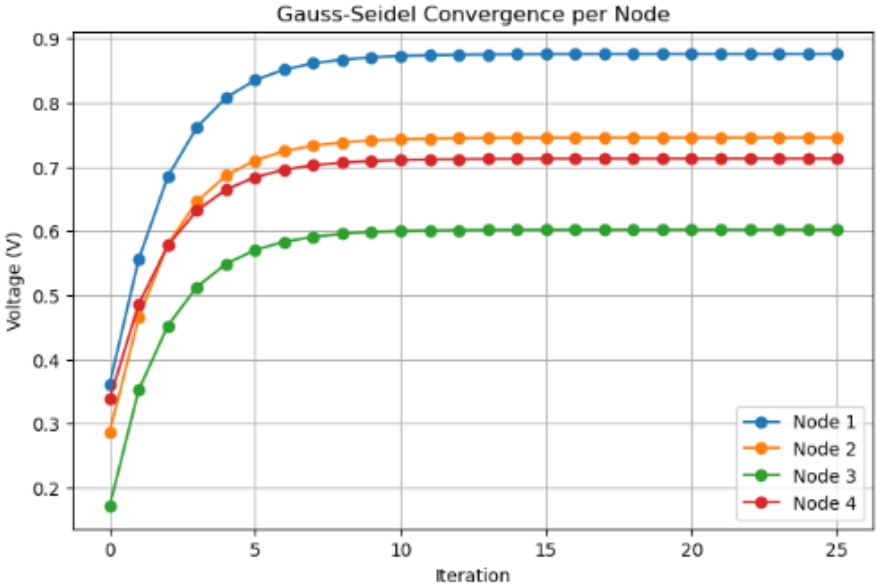

Figure 6: Gauss-Seidel Convergence Per Node

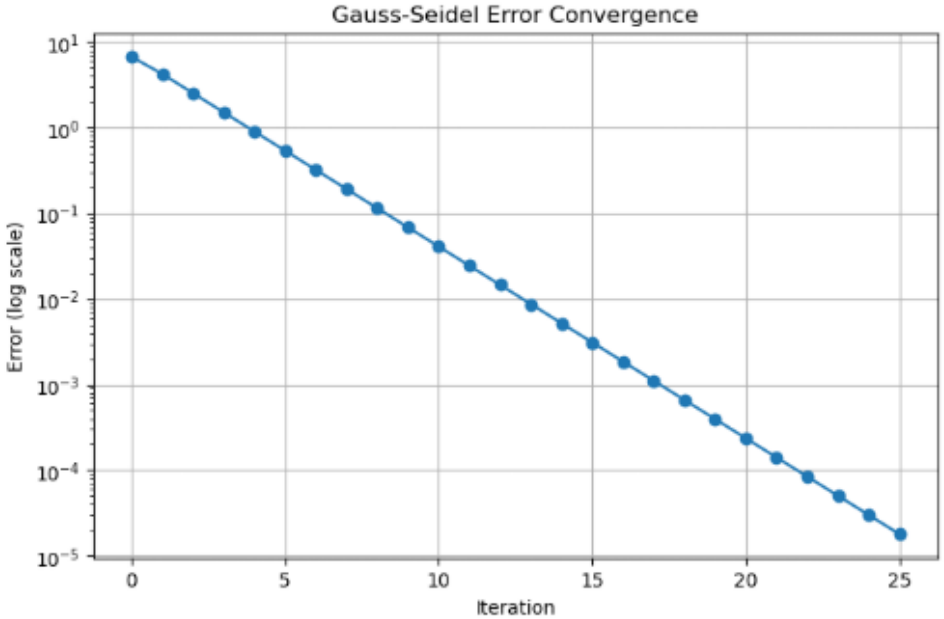

Figure 7: Gauss-Seidel Error Convergence

3.6 Comparison

Iterative methods that are used converge to voltages very close to the exact solution. SOR demonstrates the fastest convergence based on the number of its iterations, followed by Gauss–Seidel and Jacobi. The direct solver provides the reference solution with zero error. Maximum absolute errors and number of iterations for each method are summarized below.

Method	Iterations	Error
Direct Solver	N/A	0.0
SOR	21	6.82×10^{-7}
Jacobi	47	3.24×10^{-6}
Gauss-Seidel	26	1.35×10^{-6}

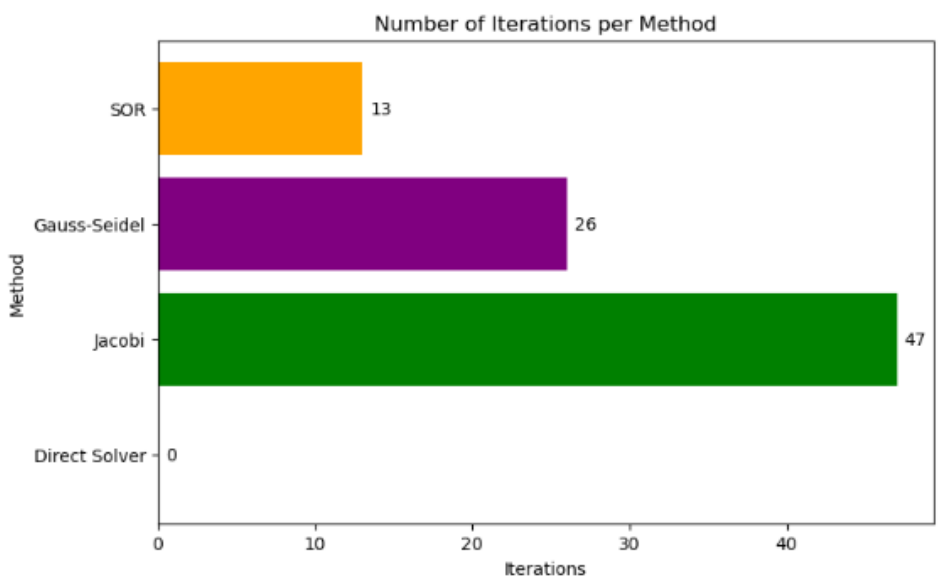

Figure 8: Number Of Iterations Per Method

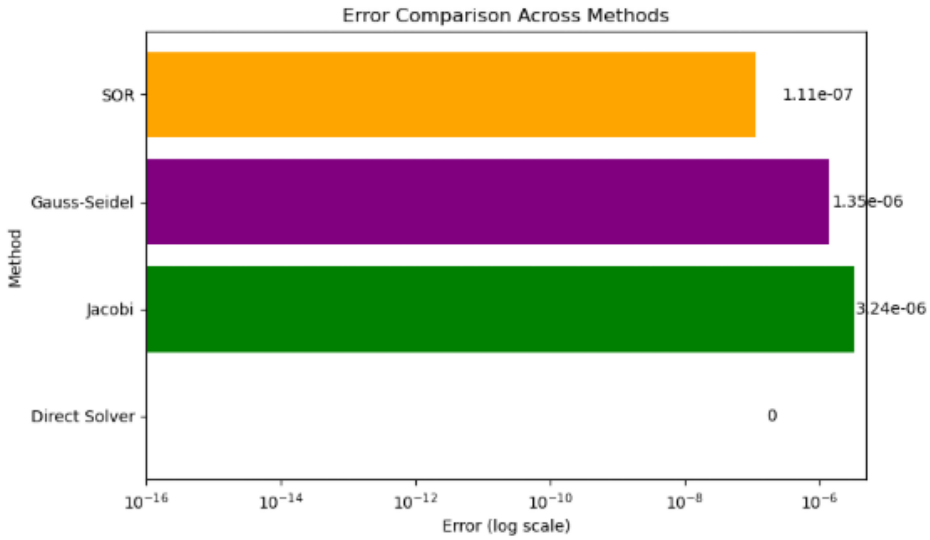

Figure 9: Error Comparison Across Method

4 Conclusion

This study successfully evaluated the performance of direct and iterative numerical methods specifically Jacobi, Gauss-Seidel, and Successive Over-Relaxation (SOR) in solving systems of linear equations derived from electrical network models using Kirchhoff's Current Law. While the direct solver method provided an exact reference solution with zero error, it is generally recognized as computationally expensive and memory-intensive for large-scale networks.

The experimental results demonstrated that all three iterative methods successfully approximated the exact node voltages, though they exhibited distinct differences in efficiency and convergence speed. The Jacobi method, which updates all variables simultaneously using only values from the previous iteration, proved to be the least efficient; it required the highest number of iterations (47) and resulted in the largest relative approximation error (3.24×10^{-6}). The Gauss-Seidel method significantly improved upon this by utilizing newly computed values immediately within the same iteration, reducing the required iterations to 26 and lowering the error to 1.35×10^{-6}.

Ultimately, the Successive Over-Relaxation (SOR) method demonstrated the most superior performance among the iterative techniques. By introducing a relaxation factor (ω) to accelerate convergence, SOR achieved the fastest convergence rate and the highest accuracy. The experimental data showed that tuning the relaxation factor (such as using $\omega = 1.25$) optimized the algorithm to converge in as few as 13 iterations with an exceptional error margin of just 1.11×10^{-7}.

In conclusion, while direct solvers remain highly reliable for establishing baseline accuracy, iterative techniques like Gauss-Seidel and especially SOR offer highly efficient, memory-saving, and scalable alternatives. This makes them highly practical and necessary computational tools for analyzing modern, complex electrical networks.

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